

Call Centre Orchestration Assessment — Customer Premium Support SG/MY

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Source: four annotated screen-and-call recordings (Cases 1–4), ~36 minutes, one technical support agent, CRM + Genesys, July 2026.

Redaction note. This is the external release of an internal assessment. Agent and staff names, case and work-order numbers, end-customer names, serial and part numbers, internal hostnames and queue names have been removed or generalised. The analysis, measurements and recommendations are unchanged.

Executive summary

70.6% of recorded handle time is silent. Across 35 minutes 40 seconds of capture, only 10 minutes 29 seconds contained speech. The remaining 25 minutes 11 seconds is the agent navigating, copying, re-typing and waiting — with the customer either listening to them type or sitting on hold. In the worst case (Case 1) the ratio reaches 89%: a 74-second customer conversation wrapped in eleven minutes of system work.

Four annotated screen-and-call recordings of one technical support agent were reviewed frame-by-frame, with audio transcribed and every on-screen system identified. The recordings are deliberately labelled walkthroughs — the capture team titled each one after its own failure mode ("WO Creation but Invalid FRU", "Existing Case — Customer Chase & WO w/ no ETA"), which means the organisation already knows where the pain is. What it does not yet have is an orchestration design that removes it.

This is not an agent performance problem. It is an integration problem wearing an agent's uniform. Every one of the four cases breaks at the same architectural seam: **there is no orchestration layer between the CRM, the parts and supply systems, the L2 knowledge holders, and the customer-facing channel.** Humans are the integration bus. They carry a part number from an email into a parts search box, carry a "no records found" back into an email, carry a warranty status from a public marketing website into a case field, and carry an escalation from a case into a group chat because the case system has no path for it.

The three highest-value plays

1. Replace human-in-the-loop parts resolution with a Parts Resolution Service. Case 1 shows the full loop: L2 supplies a part number by email → agent creates a work order → part lookup returns *No Records Found* → the work order is cancelled as a duplicate → the agent drafts another email to L2 asking them to re-check. That round trip costs days of case age and repeated agent touches, and it is entirely mechanical. A service that validates a part number against the service parts catalogue *at the point the part is proposed* — not after a work order has been built on top of it — eliminates the loop rather than speeding it up.

2. Turn the "no ETA" dead end into an event subscription. Case 4 is the clearest single waste in the sample: the agent holds a customer for five minutes, queries parts availability, receives an earliest-delivery sentinel of 31.12.2099 with a comment of *Out of Stock*, and is left with nothing to tell the customer except a promise to call

back in one to two hours after asking a dispatcher by chat. The system already knows the answer is "unknown". It should say so instantly, register the customer for notification, and push an update when supply is allocated — turning a held call plus a manual callback into zero agent touches.

3. Collapse the case-notes pipeline. The agent maintains structured call notes in **plain text files on the desktop** (a call-notes template, an escalation template, and a per-case scratch file — 3,522 characters in one observed instance), then re-types that content into the CRM Notes field, and separately into escalation emails and chat messages. Meanwhile the voice bot is *already generating* an accurate structured summary of the call intent and attaching it to the phone-call record, and an "Agent Assist & NBA" panel sits on screen reporting "*No recommendation at this time.*" The raw material for automated documentation is present and unused.

Together these three, plus the smaller items in the opportunity register below, support a realistic **35–45% reduction in after-call work and 20–30% reduction in AHT** on the case types observed, without changing a single customer-facing promise.

Evidence base

Four 1080p screen recordings with call audio, all captured on one day during a video session with roughly 23 participants — a structured walkthrough, not ambient monitoring. Each is titled on screen with its scenario and carries yellow callout annotations marking the friction points. Total runtime 35 min 40 s.

Audio was transcribed in full; 428 frames were sampled at 5-second intervals, de-duplicated to 163 distinct screen states and read individually.

#	Scenario (as labelled on screen)	Length	Speech	Silent	Silent %
Case 1	WO Creation but Invalid FRU	12:26	1:20	11:06	89.2%
Case 2	New Call & Case Creation	9:29	3:46	5:43	60.3%
Case 3	Existing Case — Troubleshooting	4:30	3:01	1:29	32.9%
Case 4	Existing Case — Customer Chase & WO w/ no ETA	9:15	2:22	6:53	74.4%
Total		35:40	10:29	25:11	70.6%

Silence was measured at a –40 dB threshold over a 1.5-second window, so it is conservative — it counts only genuine dead air, not slow speech. In Case 4, five of the six silent minutes are a single hold period while the customer waits.

Scope of the sample

Four cases handled by one agent on one day, across two country queues. Case subjects spanned a Bluetooth headset not detected, a USB-C port failure, a charger thermal incident, and a no-power fault. L2 and dispatch correspondence involves at least eight named staff across two country teams. *Agent, staff, end-customer and case identifiers have been removed from this release.*

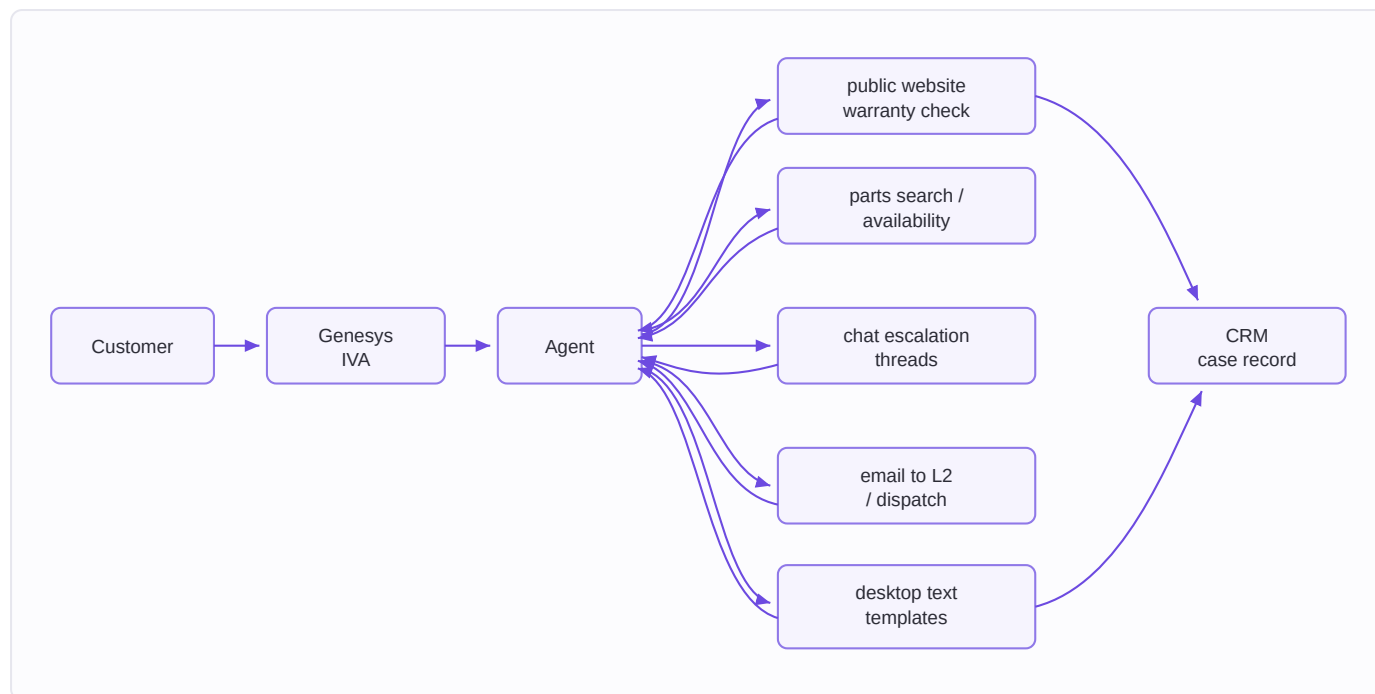
Everything below is a structural reading of what those four expose, not a statistically weighted sample. See the final section for what this cannot tell us.

Current-state system landscape

Thirteen distinct surfaces were observed in use across four calls. Eight of them are load-bearing on a single case.

Surface	What it is	Role in the observed workflow	Integration state
CRM (Dynamics 365)	Customer service hub	System of record for Case, Work Order, Contact, Customer Asset, Phone Call	Hub, but not an orchestrator
Genesys agent workspace	Softphone / interaction desktop	Call control, dispositions, outbound dialling	Screen-pop into CRM works; no post-call write-back
Genesys IVA	Voice bot front end	Captures intent, generates a case summary, attempts serial capture	Summary lands on the phone-call record and is then ignored
Agent Assist & NBA panel	In-CRM assist widget	Next-best-action suggestions	Present but inert — <i>"No recommendation at this time"</i> in every observed call
Service parts search	CRM web resource popup	Part number → part validation	Manual; returns <i>No Records Found</i> with no diagnostic
Parts availability check	CRM web resource popup	Stock and earliest-delivery lookup	Manual; returns a 31.12.2099 sentinel for unknown
Parts ordering / quoting module	Parts ordering and quoting	Chargeable-part path	Separate tab set inside the case form
Public marketing & support website	Public web	Serial → warranty status, service level, part availability	Agent uses the public website as a lookup API
Team chat	Chat	Real-time L2 and dispatcher escalation	Standing group chats named after failure modes
Email	Email	Formal L2 escalation and part requests	8–10 recipients per thread, days of latency
Desktop text editor	Text editor	Call-note templates, escalation templates, per-case scratch	Shadow system of record
Document library	Document library	Scheduling, procedure references	Separate browser tab
Remote support tool	Remote support	Remote diagnostics	Ribbon buttons, session data not linked back

How work actually flows today



Every arrow into and out of the agent is a manual transcription. The agent is the only component that talks to all the others.

The one exception: the voice platform → CRM link is genuinely integrated

One seam in the estate is not held together by a human, and it matters because it changes what the recommendations are. A real CTI and IVA integration writes structured Genesys data into the CRM on every call. The Phone Call form in Case 3 carries two dedicated read-only sections populated from the contact centre platform.

Section	Field	Observed value (Case 3)
Genesys information	Target queue	<i>[queue name redacted — encodes site, language, program and channel]</i>
Genesys information	Interaction ID	<i>[redacted]</i>
Genesys information	Serial number	<i>[redacted]</i>
Genesys information	GME Id	<i>(empty)</i>
IVA information	Is from IVA	Yes
IVA information	IVA intention	order_status, hardware_repair
IVA information	VCID	<i>[redacted]</i>
IVA information	Conversation result	Transferred - Technical Issue Beyo...
IVA information	IVA summary	A generated paragraph correctly identifying the thermal incident as a safety-related hardware issue under warranty

Supporting evidence across the other recordings:

- **Screen-pop with contact resolution.** Phone Call records are created automatically and titled by match outcome — *"Existing Contact - Existing Case"* (Case 3), *"Contact Found - New Case"* (Case 2). `Call From` is pre-bound to the resolved contact, `Call To` to the agent, and `Regarding` to the case.
- **Channel provenance flows through.** One case carries `Incoming Channel = IVA`; others show `Telephone Call` or `e-Ticketing`. Country, program, phone number and direction are all pre-filled.
- **Live transcript inside the CRM.** The Agent Assistant panel streams the conversation, timestamped, alongside IVA information, customer information and Agent Assist & NBA.
- **The Genesys workspace is separately addressable**, showing call origin, connected state and dispositions.

Current state assessment. The pipe is built, it is carrying accurate structured data, and then almost none of it is consumed. In Case 2 the IVA had already asked for the serial number — and the agent still read it back phonetically over the phone, letter by letter. The panel that should be acting on this data reports `Customer Intent: User intent not recognized yet` and `Suggestion Solution: No recommendation at this time` in every observed call. The serial number information block shows `SN is mandatory for PD` and `Please fill in the SN information and confirm` while a serial sits populated in the Genesys section three inches to its left.

Two structural limits are also visible: the flow is **Genesys → CRM only** (no disposition, outcome or resolution written back), and the IVA payload lands on the *Phone Call* record rather than being promoted onto the *Case*, which is where the agent actually works.

Proposed — what to do with this

Gap	Proposal	Effort
IVA data lands on Phone Call, not Case	Field-map promotion at screen-pop: serial, intent, contact, callback preference, IVA summary written onto the Case; agent confirms rather than re-collects	S — configuration
Serial captured but re-requested	Bind the Genesys <code>Serial Number</code> to the serial number information block and auto-resolve the asset; fall back to voice capture only on mismatch	S
<code>IVA Intention</code> unused	Route <code>IVA Intention</code> values (<code>order_status</code> , <code>hardware_repair</code>) into case type defaulting and NBA selection	S
Transcript displayed but not processed	Feed the streamed transcript into note generation (see O4) rather than only rendering it	M
No write-back to Genesys	Post-call sync of disposition, case number and outcome so routing and containment reporting reflect what actually happened	M
<code>Conversation Result</code> not acted on	Treat <code>Transferred - Technical Issue Beyond...</code> as a containment-failure signal feeding IVA tuning	S

This is the cheapest work in the entire report. **O7 and O11 are not integration builds — they are activation of an integration that already exists and already works.** That should change their position in any funding conversation.

The tell: chats named after defects

The chat panel shows standing group conversations titled **"WO Creation but Invalid FRU"**, **"Existing Case — Customer Chase & WO w/ no ETA"**, and **"Update Case while Customer OnHold"**. A chat channel named after a specific system failure mode is an organisation building a human workaround around a missing integration and

then institutionalising it. These three names are, in effect, a prioritised backlog written by the people who live with the problem.

Case-by-case walkthrough

Case 1 — "WO Creation but Invalid FRU" (12:26, 89% silent)

A Bluetooth headset that cannot be detected. Case open four days, 55 hours in progress.

The customer conversation lasts 74 seconds and consists entirely of confirming a delivery address. Everything else is the agent trying to make a part number work.

1. **00:00–03:00** — Agent reviews a case whose notes already contain a full troubleshooting history typed by a previous agent, including a hand-pasted public product URL used to find the part number.
2. **03:00** — Outbound call placed via the Genesys workspace. Address confirmed. Call ends at 04:20.
3. **06:40** — Agent opens an email thread: L2 replied several days earlier with a proposed part number. The thread has ten recipients across two country teams.
4. **07:05** — Agent opens parts search, enters the proposed part number. Result: **No Records Found**.
5. **07:45** — A work order had already been created on that part. Agent opens the work order line items cancel page and cancels it, reason *Duplicated WO*.
6. **09:00–12:26** — Agent drafts a fresh email back to L2: *"Please note that we are unable to find the required part under the provided FRU. Could you please check and let us know?"* A CRM error appears mid-draft: *This operation failed because we couldn't connect to the server.*

Orchestration gap. The part number is a data item that travels by email between two humans and is validated only after a work order has been built on it. There is no contract between the entitlement/part-recommendation step and the parts catalogue. The case will now age another cycle waiting on an email reply.

What should happen: L2's part proposal is submitted through a form or API that validates against the catalogue for the ship-to country *synchronously*. An invalid part is rejected at source with the reason (not in catalogue / not available in region / superseded by X). No work order is ever created on an unvalidated part. Case-age impact is measured in days, not minutes.

Case 2 — "New Call & Case Creation" (9:29, 60% silent)

An IT manager calling on behalf of an end user about a failing USB-C port.

The IVA has already done real work: it captured intent, produced a summary, and asked for the serial number. The screen-pop reads *Contact Found — New Case*. Then the agent discards almost all of it.

1. **00:23–01:14** — Agent re-asks for the serial number, then reads it back phonetically, then does the same for the email address. **51 seconds of call time spent on data the IVA and the CRM contact record already hold.**
2. **00:35–01:10** — Agent leaves the CRM, opens the public website, pastes the serial into the public product page to read warranty status and service level. This happens twice (wrong model first). The CRM's own *Check Entitlement* button sits unused in the ribbon.
3. **01:35–03:10** — Structured troubleshooting: BIOS updated? drivers? hardware upgrade? — read aloud from a desktop template and typed back into the CRM.
4. **03:12** — Diagnosis: physical port damage. Agent must escalate to L2 for a warranty-vs-customer-induced-damage ruling, and asks the customer to email photographs. *"I will send you an email from your ticket... just provide me with the images so I can check."*

5. Remainder — case record built by hand: title, type, contact cards, checklists, notes.

Orchestration gap. Three separate ones. (a) IVA-captured data is not promoted into the case, so the agent re-collects it by voice. (b) Entitlement is resolved from a public marketing site rather than the entitlement service behind the *Check Entitlement* button. (c) Evidence collection (photos) is an unstructured email round trip with no state — nothing knows the case is waiting on an attachment.

Case 3 — "Existing Case — Troubleshooting" (4:30, 33% silent)

A charger that overheated and caught fire.

This is the most efficient call in the sample and also the most concerning. The IVA summary on the phone-call record reads, accurately: *"The caller is following up on an existing case regarding a charger that caught fire, which is a safety-related hardware issue."* The case *Reported Symptom* field is set to a safety symptom code covering burn, smoke, swelling, melt, shock and injury.

Despite that, the case proceeds on the standard service path. The agent's own notes say *"dispatch only AC adapter after raising safety escalation"* and *"when customer calls back gather all info and escalate to L2"* — both manual, both dependent on the agent remembering. The customer, who is on an accelerator programme and needs the laptop, is told replacement takes two to three business days subject to part availability, and asks whether he can buy one outside warranty.

Orchestration gap. A structured safety symptom code is selected in the system of record and triggers nothing. No automatic severity elevation, no product-safety notification, no parts hold, no SLA override. The control exists as a dropdown value and a line in an agent's personal notes.

This is the highest-priority item in the report and it is not primarily a labour-savings item. Safety-coded symptoms should fire a deterministic workflow the moment the code is set.

Case 4 — "Existing Case — Customer Chase & WO w/ no ETA" (9:15, 74% silent)

A student needs a laptop repaired before a school deadline. He has already been promised an onsite engineer and is calling to chase.

1. **00:45** — Agent explains the alternative (service centre) costs one to two weeks. Customer stays on the dispatch path.
2. **01:34** — Address confirmed by reading it aloud from the record.
3. **02:00** — A blocking CRM validation fires: *"Current contact record is not flagged to Clean, please correct it firstly."* The agent must leave the case, open the contact record, fix a data-quality flag, and return — **mid-call, with the customer listening.**
4. **03:00** — *"Can I please put you on hold for two to three minutes while I create the repair order for you... so I can get an estimated time frame on the repair."*
5. **03:00–08:02** — **Five minutes of hold.** Agent creates a work order, opens the parts availability check, and receives: requested delivery in two days, **earliest delivery 31.12.2099**, comments **Out of Stock**. Meanwhile the agent is typing into the chat thread literally named *"Existing Case — Customer Chase & WO w/ no ETA."*
6. **08:07** — *"Currently at the moment, I don't have an estimated time for you. I will escalate your case to my dispatcher team and I will try to update you within the next one to two hours... I will give you a call back around 1pm or 2pm latest."*

Orchestration gap. The **31.12.2099** sentinel is a machine-readable statement that the supply date is unknown. It reaches a human as a date field, is interpreted by a human over chat, and converted by a human into a promise

of another human callback. The agent dashboard visible in Case 3 shows **60 open escalations** queued in exactly this pattern, aged 1.5 to 4.4 hours.

What should happen: availability returns "unknown" → the orchestration layer instantly (a) tells the agent the honest answer while the customer is still on the line, no hold; (b) opens a supply-watch subscription on that part and case; (c) routes the demand signal to planning as structured data, not chat; (d) notifies the customer by their preferred channel the moment an ETA exists. The call ends at 3:00 instead of 9:15, and the 1–2 hour callback never needs to happen.

Cross-cutting failure patterns

Seven anti-patterns recur across all four cases. Each one is a missing integration, not a missing procedure.

- 1. The human integration bus.** No system talks to another system about a specific case. Part numbers, part availability, warranty status, escalation context and customer commitments all move between systems inside an agent's short-term memory and clipboard. Every hop is a latency point and an error point.
- 2. Chat and email as the escalation transport.** L2 rulings, dispatcher ETAs and safety escalations travel as prose in chat messages and ten-recipient email threads. None of it is structured, queryable, SLA-tracked or attached to the case as state. The CRM has an *Escalations* tab; the escalation lives in chat.
- 3. Shadow system of record in a text editor.** Four text files open simultaneously, one holding 3,522 characters of case notes. Templates for call notes and escalations live on the agent's desktop. This means note structure varies by agent, nothing is analysable downstream, and the same content is typed two to four times per case.
- 4. Re-collection of data the system already has.** The IVA captures intent and attempts serial capture; the contact record holds the email and address; the asset record holds the serial and model. The agent still reads all of it back aloud, phonetically, on the call. Roughly **40–60 seconds per call** of pure re-verification.
- 5. Public website used as an internal API.** Entitlement and warranty status are read from customer-facing product pages rather than the entitlement service. This is slower, exposes marketing content and upsell banners mid-call, and produces no auditable entitlement decision on the case.
- 6. Validation gates that fire too late and in the wrong place.** *"Current contact record is not flagged to Clean"* blocks case progress mid-call. Part lookup → *No Records Found* fires after a work order exists. `31.12.2099` is a sentinel the system understands and the workflow does not. Validation is positioned to interrupt the agent rather than to prevent the bad state.
- 7. Assist infrastructure present but inert.** The IVA produces a good structured summary. The Agent Assist & NBA panel is on screen. `Suggestion Solution: No recommendation at this time` appears in every observed call. The runtime is deployed; the content and the wiring are not. This is the cheapest capability in the estate to switch on.

Where the 25 minutes of silence actually goes

Activity	Approx. share of silent time	Automatable
Waiting on / working around parts and supply answers	30%	Fully — event-driven
Typing and re-typing case notes across surfaces	25%	Largely — generated draft + review
Navigating between systems and tabs	15%	Largely — unified case context

Manual entitlement / warranty / serial lookup	12%	Fully — service call
Chasing L2 and dispatch over chat and email	10%	Largely — structured escalation
Fixing data-quality gates mid-flow	5%	Fully — pre-validate
Rework (cancelling invalid work orders)	3%	Fully — prevent at source

Shares are estimated from the frame-level screen-state analysis across the four recordings, not instrumented telemetry. They are directional and should be confirmed with desktop analytics before they are used in a funding case.

Opportunity register

Thirteen opportunities, ranked by value density (labour impact ÷ effort), with the safety item pulled to the top regardless of its labour score.

#	Opportunity	Evidence	Effort	Labour impact	Dependency
O1	Safety symptom → deterministic workflow. Safety-coded symptom fires severity elevation, product-safety notification, parts hold and SLA override automatically	Case 3	S	Low (risk, not labour)	CRM workflow only
O2	Parts availability "unknown" → supply-watch subscription. 31.12.2099 handled as an event: instant honest answer, customer notification subscription, structured demand signal to planning	Case 4, 5 min hold + 60 queued escalations	M	Very high	Availability event feed, notification channel
O3	Part validation at proposal time. L2 submits the part number through a validated form/API; invalid parts rejected at source with reason	Case 1, full rework loop	M	Very high	Parts catalogue API, L2 process change
O4	Generated case notes from IVA summary + call transcript. Draft note assembled to the standard template, agent reviews and commits	Cases 1–4, shadow text records	M	Very high	Transcript access, note schema
O5	Retire the desktop templates. Template library becomes CRM-native structured forms (call notes, escalation, onsite checklist)	All four cases	S	High	Note schema agreed first
O6	Entitlement service call replaces the public-site lookup. Wire <i>Check Entitlement</i> to return warranty status, service level and part eligibility in-case	Case 2	S	High	Entitlement API already exists behind the button

O7	IVA-to-case data promotion. Serial, intent, contact, callback preference written from the IVA payload into case fields; agent confirms rather than re-collects	Cases 2–3, 40–60 s/call	S	High	Genesys → CRM field mapping
O8	Structured escalation objects replace chat and email. Escalation becomes a first-class record with type, SLA, assignee, state; chat becomes a notification surface, not the transport	Cases 1, 3, 4	L	High	Change management with L2 and dispatch
O9	Pre-flight data-quality validation. Contact "Clean" flag and address validation resolved before the interaction, not mid-call	Case 4	S	Medium	Batch job + screen-pop check
O10	Evidence collection as a tracked state. Photo/proof-of-purchase requests issue a secure upload link; case enters <i>Awaiting Customer Evidence</i> ; auto-chase and auto-resume on receipt	Cases 2–3	M	Medium	Upload service, case state model
O11	Activate Agent Assist / NBA. Populate the inert panel with entitlement-aware, symptom-aware next actions and known-issue matches	All four — "No recommendation at this time"	M	Medium	Depends on O6, O7
O12	Proactive status push to customers. Case-state changes notify the customer on their recorded preferred channel, removing chase calls	Case 4 is a chase call that should not exist	M	High (volume deflection)	Notification platform, O2
O13	ITEM ontology as the Agent Assist knowledge layer. Asset, entitlement, symptom and part relationships made queryable in one hop; note templates become projections of the symptom node	All four — every manual lookup is a missing relation	M–L (staged — framework exists, instance is new)	Very high, compounding	Enables O4, O11; needs O6, O7 first

Effort: S = weeks within existing platform configuration · M = one to two quarters of integration build · L = multi-quarter, process and org change.

Sequencing logic

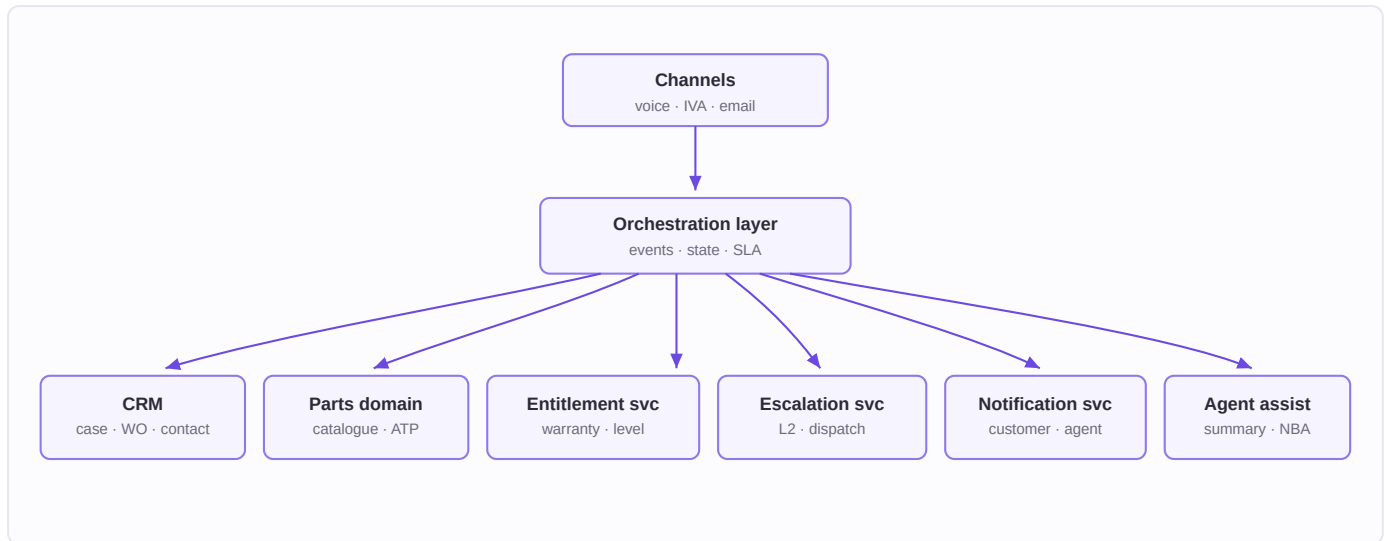
O6, O7 and O9 are configuration-weight items that make the agent's first ninety seconds honest. O4 and O5 remove the shadow record. O2 and O3 are the two structural integrations that unlock the largest blocks of silent time, and both depend on getting a real API contract with the parts and supply estate rather than a screen. O8 and O12 are the organisational plays and should not be started until O2 has proven the event model works.

O13 sits underneath O4 and O11 rather than beside them: it is the layer that makes generated notes and a working NBA possible at all. It is staged so that its first two stages ship inside the Now and Next phases, not held back as a single large programme — see *ITEM ontology as the knowledge layer for Agent Assist* below.

Target architecture

The design principle is narrow: **move integration responsibility out of the agent and into a service tier, and make case progression event-driven rather than poll-driven**. Nothing below requires replacing the CRM or the contact centre platform.

Layer model



System-of-record boundaries

These are currently blurred, which is why data is copied by hand. Making them explicit is a prerequisite to any automation.

Domain	System of record	Everyone else holds	Today's violation
Case, work order, contact	CRM	A reference	Case notes live in a desktop text editor
Part identity, part validity	Parts catalogue	A validated reference	Part numbers travel by email as free text
Part supply and ETA	Availability / planning	A subscription, never a cached date	ETA re-queried by hand on every chase call
Entitlement and warranty	Entitlement service	A decision stamped on the case	Read from the public website
Escalation state	Escalation service	A notification	Lives in chat threads
Interaction transcript and intent	Genesys	A promoted summary on the case	Discarded, then re-collected by voice

The two events that carry most of the value

`part.supply.unknown` — emitted when availability returns a sentinel or out-of-stock. Consumers: the agent desktop (show the honest answer now, suppress the hold), the escalation service (open a dispatch escalation with SLA, no chat message), planning (structured demand signal), and the notification service (subscribe this case to `part.supply.allocated`).

`part.supply.allocated` — emitted when stock or a delivery date appears. Consumers: the case (state advance, ETA stamped), the customer (proactive notification on the recorded preferred channel), and the escalation (auto-close). Case 4's entire second half, and a meaningful share of the 60 queued escalations, disappear into these two events.

Escalation as a first-class object

The chat threads are doing the job of a workflow engine. Replacing them means an escalation record with a type (`part_invalid`, `no_eta`, `safety`, `warranty_ruling`), a target queue, an SLA, a state machine and a structured payload — with chat retained purely as the notification surface. That preserves the responsiveness people currently value about chat while making the work measurable. Without this, O2 and O3 have nowhere to send their output.

Agent assist runtime

The panel already exists. The wiring it needs, in order of cost:

1. **Promote what the IVA already produced** — intent, summary, serial, contact, callback preference — into case fields at screen-pop. No model required.
2. **Stamp the entitlement decision** on the case from the entitlement service, replacing the public-website trip.
3. **Generate the case note draft** from the IVA summary plus the call transcript, rendered into the agreed note template for agent review and commit. This is where the shadow record dies.
4. **Populate NBA** with known-issue matches, symptom-to-part mappings, and entitlement-aware next actions.

Steps 1 and 2 are integration work with no AI dependency and should not wait for step 3.

Deliberate non-goals

- **Do not automate the customer conversation further.** The talk time in these recordings is already efficient; the waste is entirely on the system side. Adding containment pressure to the IVA would push cost into repeat contacts.
- **Do not build a desktop-automation (RPA) bridge across these gaps.** It would encode the current screen layouts as a dependency and make the real integrations harder to justify later. The one defensible RPA use is a short-lived shim in front of a system with a firm API roadmap.
- **Do not consolidate the CRM and the Genesys desktop into a single pane as the first move.** A unified pane over unintegrated back ends just relocates the copying.

ITEM ontology as the knowledge layer for Agent Assist

What is confirmed and what is assumed. The framework exists in the estate today: an `OntologyExecutor` MCP connector is installed at org level, and the `itemGPT` agent runtime serves natural-language queries over structured WMS/TMS business data. A customer deployment would be a **separate instance of the same framework**, not a new build — which changes this from a platform question to a provisioning and modelling one. What could not be verified here is the ontology schema itself: `OntologyExecutor` was in an incomplete connection state and was not queryable from this session. The four entity classes below are therefore derived from what the four recordings demanded, and should be reconciled against the actual ITEM schema as the first task.

Why the current Agent Assist cannot work

The panel is not failing because the model is weak. It is failing because it has **no addressable world**. Everything it would need to reason over is either locked inside a form field, sitting in a text file on the agent's desktop, or written as prose in a chat message. `Customer Intent: User intent not recognized yet` is what an assist runtime says when it can see a transcript and nothing else.

An ontology changes the question the runtime is being asked. Today it is asked *"what should this agent say next?"* with only the transcript as input. With an ontology it is asked *"given this asset, this entitlement, this symptom, this part, this supply state and these forty prior resolutions of the same shape — what is the next action?"* That is a retrieval problem with a defensible answer, not a generation problem with a guess.

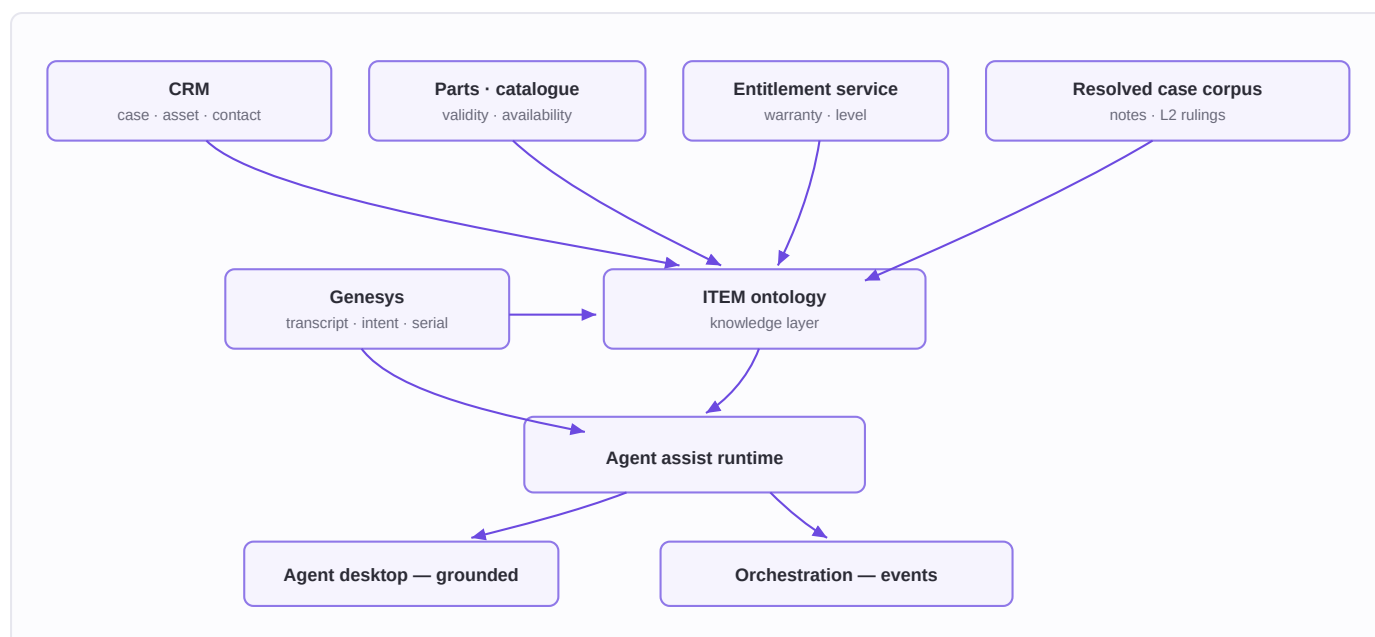
The four entity classes the recordings demand

Each of these is drawn from something the agent had to hold in their head or look up manually across the four cases.

Entity class	Nodes	Relations the recordings needed	Case evidence
Asset & product	Serial, machine type model, product family, component	<code>serial → has_model</code> , <code>model → has_component</code> , <code>component → serviced_by_part</code>	Case 2: <code>serial → model → USB-C port → planar part</code>
Entitlement & service	Warranty term, service level, proof of purchase, programme, region	<code>asset → entitled_to</code> , <code>entitlement → permits_service_type</code> , <code>region → constrains_fulfilment</code>	Case 2: agent read this off the public website instead
Symptom & resolution	Symptom code, diagnostic step, resolution, damage- cause ruling	<code>symptom → requires_check</code> , <code>symptom → resolved_by</code> , <code>symptom → escalates_as</code>	Case 3: the safety symptom code triggered nothing
Part & supply	Part number, catalogue validity, availability state, vendor, collection location	<code>part → valid_for_model_in_region</code> , <code>part → has_supply_state</code> , <code>part → substitutable_by</code>	Case 1: invalid part; Case 4: <code>31.12.2099</code> out of stock

The fourth row is the one that would have prevented Case 1 entirely. `part → valid_for_model_in_region(SG)` is a single ontology assertion. It was instead resolved by a work order, a failed part search, a cancellation and an email thread spanning several days.

How it fits the estate



The ontology is a **read model**, not a new system of record. The CRM still owns the case; the parts catalogue still owns part identity. The ontology's job is to make the relationships between them queryable in one hop, which is precisely the work the agent is doing by hand today.

What the agent experience becomes

Taking Case 2 — the USB-C port — as the worked example.

Moment	Today	With the ontology behind Agent Assist
Call connects	Screen-pop with contact; agent re-asks serial and reads it back phonetically (51 s)	Serial arrives from Genesys, resolves to asset, model, warranty term and service level; agent confirms in one line
Entitlement check	Agent opens the public website, pastes serial, reads warranty status off a marketing page (twice — wrong model first)	Entitlement decision already stamped on the case with its provenance; no navigation
Symptom capture	Agent reads a checklist from a desktop template and types answers into the case	Assist proposes the checklist <i>for this symptom on this model</i> , pre-filled where the transcript already answered it; agent corrects exceptions only
Known-issue match	None	USB-C port intermittent on this model matches 40 prior cases; 31 resolved by planar replacement, 6 ruled customer-induced — shown with the ruling criteria
Part identification	Manual browse of public product pages to find a part number	Part derived from model → component → part, validated against the catalogue before anything is created
Evidence request	Agent composes an email asking for photos; case state unchanged	Assist issues the request from the symptom's evidence requirement; case enters <i>Awaiting Evidence</i> with auto-chase
Case notes	Typed in a text editor, re-typed into CRM, re-typed into the escalation email	Draft note assembled from the ontology-resolved facts plus the transcript, rendered to the standard template; agent reviews and commits

The note problem, specifically

The copy-paste and typing burden is not one problem, it is three, and the ontology solves them in different ways.

1. Facts the agent re-types because they live in another pane. Serial, model, warranty, address, part number. These do not need generation at all — they need resolution. The ontology returns them as structured values that populate fields directly. This is the largest share of the typing and the easiest to remove.

2. Structure the agent re-creates from a desktop template. The text files are, in effect, an ontology the agent maintains by hand: a checklist per symptom class, an escalation format per escalation type. Modelled properly, the note template becomes a *projection of the symptom node* — the checklist for a safety symptom differs from the checklist for a Bluetooth symptom because the ontology says it does. The agent never picks a template again.

3. Narrative the agent writes from the conversation. This is the only part that genuinely needs generation, and it is the smallest part. Grounded in resolved entities rather than free text, the draft is checkable: every factual claim in the note traces to an ontology node or a transcript span, which is what makes agent review fast rather than a re-read.

The design rule: **generate only the narrative, resolve everything else.** Systems that try to generate the facts as well are the ones agents stop trusting.

Grounding and safety

Two controls are non-negotiable given what Case 3 showed.

Deterministic paths stay deterministic. Safety symptom codes, entitlement rulings and damage-cause determinations route through ontology rules, not a language model. The assist runtime may surface them and explain them; it may not decide them. A thermal incident escalates because `symptom → escalates_as(safety)` is an assertion, not an inference.

Every suggestion carries provenance. An agent who cannot see why a part was proposed will go back to the public website, and the whole investment is lost. Each assist output should name the nodes it traversed and the source records behind them — this is also what makes the L2 rework rate measurable.

Sequencing

The ontology does not need to be complete to be useful, and building it wholesale before shipping anything is the main way this fails.

Stage	Scope	Unlocks	Prerequisite
1	Asset → model → entitlement resolution	Removes serial re-collection and the public-website trip	Genesys field promotion (O7), entitlement API (O6)
2	Symptom → checklist → evidence requirement	Retires the desktop templates; drives structured capture	Symptom taxonomy agreed with L2
3	Part → catalogue validity → availability state	Kills the invalid-part loop and the no-ETA hold	Parts domain API contract (O2, O3)
4	Resolution corpus → known-issue matching	Real NBA; L2 escalation volume falls	Stages 1–3 plus back-loaded case history
5	Grounded note generation	After-call work collapses	Stage 2 note schema, transcript access

Stages 1 and 2 are worth doing on their own merits even if nothing downstream is funded. Stage 4 is where the compounding return sits, and it is also the stage that cannot be bought — it depends on the resolution corpus being structured, which is exactly what stages 2 and 5 produce. **The notes the ontology helps agents write today are the training data the ontology needs to be useful in eighteen months.** That is the argument for starting the note work early rather than treating it as a convenience feature.

Building on the existing ITEM framework

Because the framework already exists, a customer instance inherits most of the hard parts. What it needs is a domain model, not a platform.

Framework capability (already built)	What the customer instance inherits	What still has to be built
Multi-tenancy — tenant id resolved from the token, never hardcoded	Tenant isolation between existing logistics data and customer service data by construction; no data-commingling design work	Tenant provisioning, IAM integration with the customer estate
OntologyExecutor as an MCP server	The assist runtime addresses the ontology over a standard protocol — no bespoke connector per consumer; CRM, the orchestration layer and the agent desktop are all just clients	Completing the connector setup, plus the customer entity and relation definitions
itemGPT chat-completions API with agent, chat and stream parameters	A working agent surface: chat id maps cleanly to a call session so the assist panel holds context across an interaction; streaming supports live in-call rendering	A service-domain agent profile with the right prompting and tool scope
Traversal steps returned in the response	The provenance mechanism already exists — the traversal is returned alongside the answer, which is exactly what the grounding requirement above needs; it does not have to be invented	Surfacing the traversal in the agent UI as a readable "why this part / why this ruling" trail
Password-grant auth, long-lived tokens	Session-length credentials that survive a shift	Mapping customer agent identity to itemGPT principals

The traversal finding is the significant one. The grounding control described above — *every suggestion carries provenance* — is usually the expensive part of an assist deployment and the part that gets cut. Here it is already in the response contract. That should be treated as a design constraint to preserve rather than a feature to add later.

The deterministic paths — safety escalation, evidence collection, the supply-watch subscription — need an execution home distinct from the assist runtime, since they must fire on rules rather than on inference. The requirement is a workflow engine that shares the ontology's tenant scoping, supports streaming for live in-call steps, and can hold each path as a component querying the ontology as its read model. Selecting that engine is an open item and should be assessed against the requirement rather than assumed.

Open questions to settle before Stage 3

- 1. Schema fit.** Does the ITEM ontology's work-item primitive accommodate a service case with an asset, an entitlement and a part, or does the customer instance need its own entity classes? This is answerable in one session against the live schema and gates everything else.
- 2. Where part and supply data lands.** Catalogue validity and availability state are customer-side systems. Does the ontology hold them as nodes, or federate to them at query time? Federation is more honest about staleness; holding them is faster. The 31.12.2099 case argues for federation with an event subscription rather than a cached date.

3. **Tenancy and commercial model.** A customer instance running on ITEM infrastructure is a different commercial conversation from a customer-hosted deployment of the same framework. This is not a technical blocker but it will determine the delivery path, and it should be resolved before Stage 3 rather than during it.

Labour and AHT model

Stated assumptions

These are the numbers the model turns on. All four should be replaced with real operational data before this is used for funding.

- Observed handle time averages **8:55** per interaction across the four recordings, of which **6:18** is silent.
- The sample is skewed toward hard cases (that is what the walkthrough was for). Assume the true mix is **60% of the observed friction level**.
- Per-interaction after-call work is assumed at **4 minutes**, consistent with the note-typing observed in Cases 1 and 2.
- Volume, headcount and fully-loaded cost per agent-hour are **not known from this sample** and must come from the WFM system.

Bottom-up time recovery per interaction

Source	Mechanism	Est. saving	Confidence
Re-collecting serial, email, address	O7 IVA data promotion	40–60 s	High — directly timed
Entitlement lookup via public site	O6 entitlement service	30–45 s	High — directly timed
Hold time waiting on parts answers	O2 supply-watch	60–300 s on affected calls	High — directly timed
Mid-call data-quality fixes	O9 pre-flight validation	20–40 s on affected calls	Medium
Note typing and re-typing (ACW)	O4 + O5 generated notes	90–150 s	Medium
Manual escalation composition	O8 structured escalation	45–90 s on affected calls	Medium

Headline ranges

Applying the 60% haircut to the observed friction:

- **AHT reduction: 20–30%** on the case types observed — driven mainly by O2, O6 and O7.
- **After-call work reduction: 35–45%** — driven mainly by O4 and O5.
- **Contact deflection: 5–12%** of inbound volume — chase calls like Case 4 that O12's proactive notification removes entirely.
- **Rework elimination:** the invalid-part loop in Case 1 consumed at least two full agent touches plus multi-day case age. O3 removes the loop rather than shortening it, so the saving is in case age and repeat contacts, not minutes.

Multiply the AHT and ACW percentages by your own volume and cost-per-agent-hour to size the case. On any normal large-scale contact centre these percentages are the difference between needing the next hiring wave and not.

The non-financial case

Two items should not be argued on labour savings at all.

Safety (O1). A charger that overheated and caught fire was coded correctly in the system of record and still travelled the standard service path, with the safety escalation existing only as a line in an agent's personal notes. That is a control gap, and it should be funded as one.

Promise integrity (O2, O12). Case 4 ends with a commitment to call back "around 1pm or 2pm latest" that depends on a dispatcher answering a chat message. The dashboard shows 60 escalations queued in the same pattern. Every one of those is a promise whose keeping is unmanaged.

Phased roadmap

Now — 0 to 3 months: stop the bleeding, prove the model

Everything here is configuration or thin integration inside systems already licensed.

Item	Success metric
O1 Safety symptom workflow	100% of safety-coded symptoms trigger escalation without agent action
O7 IVA data promotion to case	Serial/contact re-verification time per call drops below 15 s
O6 Entitlement service replaces the public-site lookup	Zero navigations to the public site during observed calls
O9 Pre-flight contact data quality	Zero mid-call "not flagged to Clean" interruptions
O5 Retire desktop templates	Note templates are CRM-native; agent desktops carry no case data

Entry criteria: none beyond sponsor sign-off. **Exit criteria:** re-record the same four scenarios and re-measure the silent-time ratio. The target is 70.6% down to roughly 55%.

Next — 3 to 9 months: the two structural integrations

Item	Success metric
O3 Part validation at proposal	Invalid-part work orders per month → zero
O2 Parts availability event model	Hold time on parts-status calls → under 30 s; no-ETA escalation queue depth falls by half
O8 Structured escalation objects	Escalations resolvable from the case record; chat used only for notification
O4 Generated case notes	Agent edits rather than writes the note; ACW down 35%+

Entry criteria: an agreed API contract with the parts and supply domain (not a screen-scrape), and the escalation taxonomy signed off by L2 and dispatch leadership. **Exit criteria:** the `part.supply.unknown` and `part.supply.allocated` events are in production and consumed by at least three subscribers.

Later — 9 to 18 months: proactive service

Item	Success metric
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O12 Proactive customer status push	Chase-call volume down 5–12%
O10 Evidence collection as tracked state	Zero cases silently stalled awaiting customer photos
O11 Agent Assist / NBA populated	NBA suggestion accepted on a material share of interactions
Cross-region rollout	SG/MY pattern extended to other regional queues

Entry criteria: O2 event model stable in production; customer contact preferences reliable in the CRM.

Governance

One metric should be reported every month from the start, because it is the single number that tells you whether any of this is working: **silent-time ratio per interaction**, measured by desktop analytics rather than by re-recording. It is the closest available proxy for "how much of this agent's shift is spent being an integration bus." Everything in this roadmap is an attempt to move that one number.

Risks, caveats and what this sample cannot tell us

Limits of the evidence

- **Four cases, one agent, one day.** The scenarios were selected for a walkthrough, which means they are deliberately the hard ones. Every percentage in this report is directional until confirmed against operational data.
- **No volume data.** Without contact volumes by case type, the opportunity register can be ranked but not priced.
- **Screen-derived timings.** Silent-time shares come from frame sampling at 5-second intervals plus audio energy analysis. They are reliable at the level of "most of this is silent" and unreliable at the level of "exactly 30% is parts-related."
- **One agent's technique.** Some of what looks like process may be personal habit — the desktop templates in particular. Whether the whole floor works this way is the first thing to check, and it changes the sizing significantly in either direction.
- **APIs assumed, not verified.** The report assumes the parts catalogue, availability and entitlement systems can expose services rather than only web-resource pages. That assumption carries the two largest opportunities and must be validated before anything is committed.

Delivery risks

Risk	Why it matters here	Mitigation
Parts and supply systems cannot expose real APIs	O2 and O3 both depend on it; without it the value collapses to the Now phase	Validate in week one, before roadmap commitment
L2 and dispatch resist structured escalation	Chat is fast and familiar; a form feels slower to the people using it	Keep chat as the notification surface; never make the structured path slower than chat
Generated case notes degrade note quality	Downstream teams rely on these notes for L2 rulings	Agent reviews and commits; measure L2 rework rate before and after
Automation absorbs capacity rather than releasing it	Savings quietly become longer calls	Track silent-time ratio, not just AHT

Proactive notification increases inbound	Poorly worded status pushes generate questions	Pilot on one case type, measure net contact volume
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Recommended next data collection

In rough priority order:

1. **Desktop analytics across the floor for two weeks** — application dwell time, switch counts, and silent-time ratio per interaction. This converts every estimate in this report into a measured number.
2. **Volume by case type and by scenario pattern** — how many interactions per month are invalid-part loops, no-ETA chases, and new case creations.
3. **Escalation census** — pull the escalation chats and the dispatcher queue for one month and classify them. The 60 queued escalations on the observed dashboard are the sizing input for O2 and O8.
4. **API discovery session with the parts and supply domain owners** — one meeting that determines whether the Next phase is viable as written.
5. **Five more recordings from different agents** — specifically to test whether the desktop-template pattern and the public-website lookup are universal or individual.

Glossary of abbreviations

Every abbreviation used above. Several could not be expanded with confidence from the recordings alone and are marked accordingly — they are customer-internal names and should be confirmed with the account team rather than guessed at.

Abbr.	Expansion	What it means in this report
3PL	Third-Party Logistics	Outsourced logistics and fulfilment provider — used as the scale reference for the labour model
AC	Alternating Current	As in "AC adapter", the mains power brick — the failed part in Case 3
ACW	After-Call Work	Time an agent spends on a case after the customer hangs up: notes, case updates, escalations
ADP	Accidental Damage Protection	A warranty add-on; determines whether physical damage is covered or chargeable
AHT	Average Handle Time	Mean duration of an interaction including hold and after-call work; the standard contact-centre productivity measure
API	Application Programming Interface	A machine-to-machine interface, as opposed to a screen a person has to read
ASEAN	Association of Southeast Asian Nations	The regional grouping the SG/MY queues sit within; the rollout scope in the Later phase
ATP	Available to Promise	The supply check that answers "can we commit this part by this date" — the source of the 31.12.2099 sentinel in Case 4
BIOS	Basic Input/Output System	Firmware layer; "BIOS updated?" is a standard item on the troubleshooting checklist
CID	Customer-Induced Damage	A ruling that damage was caused by the customer, making the repair chargeable rather than warranty-covered

CRM	Customer Relationship Management	Here, Dynamics 365 — the system of record for cases, work orders and contacts
CTI	Computer Telephony Integration	The link between the phone platform and the CRM that produces the screen-pop
dB	Decibel	Unit of sound level; the –40 dB threshold used to distinguish speech from silence
ETA	Estimated Time of Arrival	The delivery date for a part — absent in Case 4, which is the whole problem
FCR	First Contact Resolution	Share of contacts resolved without a repeat contact
FRU	Field Replaceable Unit	A part number identifying a component a technician can swap; the item that failed validation in Case 1
IVA	Intelligent Virtual Agent	The Genesys voice bot that answers first, captures intent and serial, then transfers to a human
ITEM	<i>Product name, not an initialism</i>	ITEM's platform family — ticketing, itemGPT agent, ontology, WMS/TMS integration
JWT	JSON Web Token	The signed token itemGPT issues at login; the tenant id is read from its payload rather than hardcoded
L2	Level 2 (support)	The escalation tier holding parts, warranty and engineering rulings; reached today by email and chat
MCP	Model Context Protocol	The open protocol by which tools expose capabilities to an AI runtime; how OntologyExecutor is surfaced
MTM	Machine Type Model	The product identifier that links a serial number to a parts list
NBA	Next Best Action	The recommendation surface in the Agent Assist panel; currently returns nothing
NCA	Next Contact Avoidance	Whether an interaction prevented a follow-up contact; the routing and QA signal in the Later phase
ONS	Onsite	Case-title prefix indicating an onsite engineer visit
POP	Proof of Purchase	Receipt evidence required to establish warranty start date; requested from customers by email today
PUDO	Pick-Up / Drop-Off	A collection point used as a delivery destination for parts
QA	Quality Assurance	Interaction scoring and coaching
RPA	Robotic Process Automation	Screen-level automation that drives existing UIs; named in the report as a deliberate non-goal
SBOM	Service Bill of Materials	The parts catalogue that determines whether a part is valid for a given model and region
SG/MY	Singapore / Malaysia	The two country queues covered by the observed agent and dashboard
SLA	Service Level Agreement	The committed response or resolution time; absent from the chat-based escalation path
SN	Serial Number	The device identifier; re-collected by voice in Case 2 despite already being present
SoR	System of Record	The authoritative source for a given data domain — the boundary the report argues must be made explicit

STA	Speech & Text Analytics	Contact-reason, topic and conformance analysis over interactions
TMS	Transport Management System	Freight and transport execution system; part of the ITEM/itemGPT data domain
UI	User Interface	The screen an agent works in
WEM	Workforce Engagement Management	Coaching, learning and gamification in the contact centre platform
WFM	Workforce Management	The scheduling and forecasting system that holds the volume and headcount data the labour model needs
WMS	Warehouse Management System	Warehouse execution system; part of the itemGPT data domain
WO	Work Order	The dispatch and parts record created from a case; cancelled as a duplicate in Case 1

O1–O13 are not abbreviations but identifiers for the entries in the opportunity register above, used so that the roadmap and architecture sections can refer to them without restating them.

S / M / L in the register denote effort: weeks within existing platform configuration, one to two quarters of integration build, and multi-quarter process and organisational change respectively.

Analysis derived from four screen and call recordings supplied September 2026, captured July 2026. Audio transcribed in full; 428 frames sampled and 163 distinct screen states read. No external sources were used — all findings are from the recordings themselves.

External release. Customer, agent, end-customer, case, asset and infrastructure identifiers have been removed or generalised. Figures, measurements, findings and recommendations are unchanged from the internal version.